

Answers to the Q. No-81

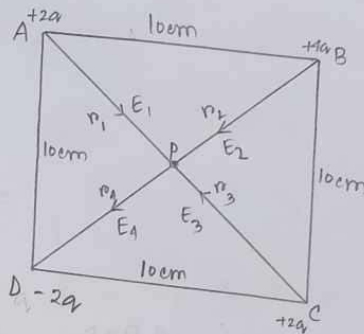
81

$$E_1 = \frac{k \cdot q_1}{r_1^2} = \frac{2kq}{r^2}$$

$$E_2 = \frac{k \cdot q_2}{r_2^2} = \frac{4kq}{r^2}$$

$$E_3 = \frac{k \cdot q_3}{r_3^2} = \frac{2kq}{r^2}$$

$$E_4 = \frac{k \cdot q_4}{r_4^2} = \frac{2kq}{r^2}$$



$$\begin{aligned} \therefore E_{13} &= E_1 - E_3 \\ &= \frac{2kq}{r^2} - \frac{2kq}{r^2} \\ &= 0 \end{aligned}$$

$$\begin{aligned} \therefore E_{24} &= E_2 + E_4 \\ &= \frac{4kq}{r^2} + \frac{2kq}{r^2} \\ &= \frac{6kq}{r^2} \end{aligned}$$

Here,

$$r_1 = r_2 = r_3 = r_4 = r$$

$$D = \sqrt{2}a = 0.1\sqrt{2}$$

$$r = \frac{0.1\sqrt{2}}{2}$$

$$= 0.0707m$$

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$$\begin{aligned}\therefore E_{\text{net}} &= E_{13} + E_{24} \\ &= 0 + \frac{6Kq}{r^2} \\ &= \frac{6 \times 9 \times 10^9 \times 57 \times 10^{-6}}{\left(\frac{0.1\sqrt{2}}{2}\right)^2} \\ &= 6.156 \times 10^8 \text{ N/C}\end{aligned}$$

$\therefore$ ABCD,

$$\begin{aligned}\cos \theta_{\text{net}} &= \frac{CD}{BD} \\ \Rightarrow \theta_{\text{net}} &= \cos^{-1} \left\{ \frac{a}{a\sqrt{2}} \right\} \\ &= \cos^{-1} \left(\frac{1}{\sqrt{2}} \right) \\ &= 45^\circ\end{aligned}$$

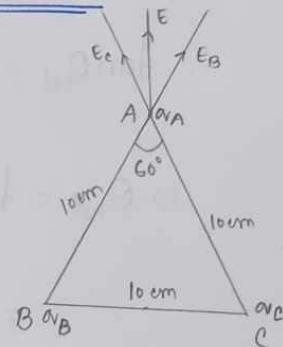
Ans.

Answers to the Q. NO-02

03

$$\begin{aligned} E_B &= k \cdot \frac{q_B}{r^2} \\ &= \frac{9 \times 10^9 \times 12 \times 10^{-7}}{(0.1)^2} \\ &= 1.08 \times 10^6 \text{ N/C} \end{aligned}$$

$$\begin{aligned} E_C &= k \cdot \frac{q_C}{r^2} \\ &= \frac{9 \times 10^9 \times 12 \times 10^{-7}}{(0.1)^2} \\ &= 1.08 \times 10^6 \text{ N/C} \end{aligned}$$



Here,
 $q_A = q_B = q_C = q = 12 \times 10^{-7} \text{ C}$
 $r = 10 \text{ cm}$
 $= 0.1 \text{ m}$

$$\begin{aligned} E &= \sqrt{E_B^2 + E_C^2 + 2E_B E_C \cos 60^\circ} \\ &= \sqrt{(1.08 \times 10^6)^2 + (1.08 \times 10^6)^2 + 2 \times (1.08 \times 10^6) \times \cos 60^\circ} \\ &= 1.87 \times 10^6 \text{ N/C} \end{aligned}$$

Teacher's Signature:

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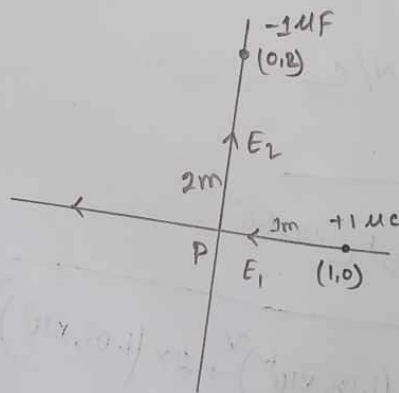
$$\therefore \tan \theta_{\text{net}} = \frac{E_B \sin 60^\circ}{E_C + E_B \cos 60^\circ}$$

$$\Rightarrow \theta_{\text{net}} = \tan^{-1} \left\{ \frac{1.08 \times 10^{-6} \times \sin 60^\circ}{1.08 \times 10^{-6} + 1.08 \times 10^{-6} \cos 60^\circ} \right\}$$

$$= 30^\circ$$

Ans.

Answers to the Q.No-03



$$E_1 = \frac{k q_1}{r_1^2}$$

$$= \frac{9 \times 10^9 \times 1 \times 10^{-6}}{1^2}$$

$$= 9000 \text{ N/C}$$

$$E_2 = \frac{K a v_2}{r_2^2}$$

$$= \frac{9 \times 10^9 \times 1 \times 10^{-6}}{(2)^2}$$

$$= 2250 \text{ N/C}$$

$$\therefore E_{\text{net}} = \sqrt{E_1^2 + E_2^2}$$

$$= 9276.98 \text{ N/C}$$

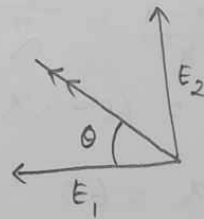
ii)

$$\tan \theta = \frac{E_2 \sin 90^\circ}{E_1 + E_2 \cos 90^\circ}$$

$$\Rightarrow \theta = \tan^{-1} \left\{ \frac{E_2}{E_1} \right\}$$

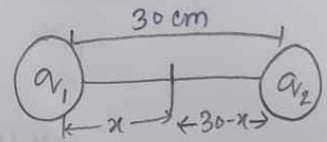
$$= 14.03^\circ$$

$\therefore \theta$ is 14.03° above (-) x-axis clock wise.



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Answers to the Q. No-04



$$\therefore E_1 = E_2$$

$$\Rightarrow \frac{k \cdot q_1}{x^2} = \frac{k \cdot q_2}{(30-x)^2}$$

$$\Rightarrow \frac{q_1}{x^2} = \frac{q_2}{(30-x)^2}$$

$$\Rightarrow \frac{(30-x)^2}{x^2} = \frac{q_2}{q_1}$$

$$\Rightarrow \frac{30-x}{x} = \sqrt{\frac{q_2}{q_1}}$$

$$\Rightarrow \frac{30-x}{x} = \frac{1}{2}$$

$$\Rightarrow x = 60 - 2x$$

$$\Rightarrow 3x = 60$$

$$\Rightarrow x = 60/3$$

$$\therefore x = 20 \text{ cm}$$

Here,

$$q_1 = 4q$$

$$q_2 = +q$$

Answer to the Q. No-05

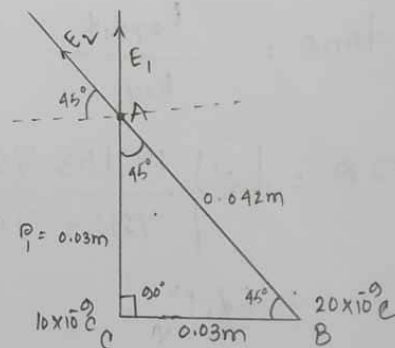
Since dipole is placed in a uniform electric field. So, it experiences zero external force. Also, dipole is placed to parallel to electric field angle between $\vec{p}$ and $\vec{E}$ is zero. $\theta = 0^\circ$

$$\begin{aligned}\text{Torque, } \tau &= \vec{p} \times \vec{E} \\ &= pE \sin \theta \\ &= pE \sin 0^\circ \\ &= 0 \text{ Nm}\end{aligned}$$

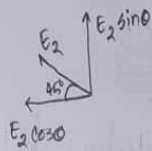
Answer to the Q. No-06

$$\begin{aligned}E_1 &= \frac{k q_C}{r_1^2} \\ &= \frac{9 \times 10^9 \times 10 \times 10^{-9}}{(0.03)^2} \\ &= 1 \times 10^5 \text{ N/C}\end{aligned}$$

$$\begin{aligned}E_2 &= \frac{k q_B}{r_2^2} = \frac{9 \times 10^9 \times 20 \times 10^{-9}}{(0.042)^2} \\ &= 1.02 \times 10^5 \text{ N/C}\end{aligned}$$



$$\begin{aligned}r &= \sqrt{(0.03)^2 + (0.03)^2} \\ &= 0.042 \text{ m}\end{aligned}$$



$$\theta = \cos^{-1} \left(\frac{0.03}{0.042} \right) = 45^\circ$$

$$E_{2x} = E_2 \cos \theta = 72153.74 \text{ N/C}$$

$$E_{2y} = E_2 \sin \theta = 72153.74 \text{ N/C}$$

$$\therefore E_{2y \text{ net}} = E_1 + E_{2y} = 1 \times 10^5 + 72153.74 = 172153.74 \text{ N/C}$$

$$\therefore E_{2 \text{ net}} = \sqrt{E_2^2 + E_{2y \text{ net}}^2} = 186662.99 \text{ N/C}$$

$$\tan \theta = \frac{E_{2y \text{ net}}}{E_{2x}}$$

$$\Rightarrow \theta = \tan^{-1} \left\{ \frac{172153.74}{72153.74} \right\} = 67.1^\circ$$

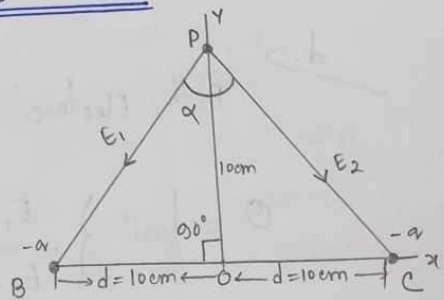
θ is 67.1° above $(-)$ x axis. from positive x axis it will be 150°

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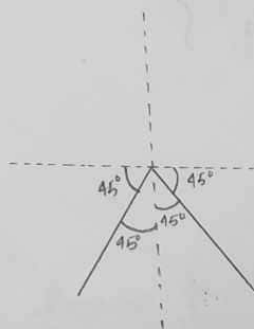
Answers to the Q. No-07

a.) $E_1 = \frac{kq_1}{(0.14)^2} = \frac{9 \times 10^9 \times 10 \times 10^{-5}}{(0.14)^2}$
 $= 4.5 \times 10^7 \text{ N/C}$

$E_2 = \frac{kq_2}{(0.14)^2} = \frac{9 \times 10^9 \times 10 \times 10^{-5}}{(0.14)^2}$
 $= 4.5 \times 10^7 \text{ N/C}$



b.)



$\theta = \tan^{-1} \left\{ \frac{0.1}{0.1} \right\}$
 $= 45^\circ$

4PBC,

$PB^2 = PO^2 + BO^2$

$\Rightarrow PB = \sqrt{10^2 + 10^2}$
 $= 10\sqrt{2} \text{ cm}$

$\therefore PB = PC = r = 10\sqrt{2} \text{ cm}$
 $= 0.1414 \text{ m}$

$q = 10 \times 10^{-5} \text{ C}$

c.)

$E_{\text{net}} = \sqrt{E_1^2 + E_2^2 + 2E_1E_2 \cos 90^\circ}$
 $= \sqrt{(4.5 \times 10^7)^2 + (4.5 \times 10^7)^2 + 2 \times (4.5 \times 10^7)^2 \cos 90^\circ}$
 $= 6.36 \times 10^7 \text{ N/C}$

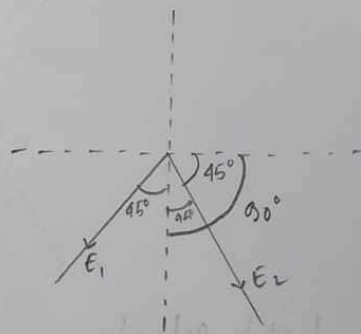
d. → Both Electric field magnitude is same.

$$\theta = \tan^{-1} \left\{ \frac{E_1 \sin 90^\circ}{E_2 + E_1 \cos 90^\circ} \right\}$$

$$= \tan^{-1} \left\{ \frac{E_1}{E_2} \right\}$$

$$= 45^\circ$$

E_{net} is 45° from E_2



e. → X component:

$$E_{1x} = E_1 \cos \theta$$

$$= 4.5 \times 10^7 \cos 45^\circ$$

$$= 3.18 \times 10^7 \text{ N/C}$$

$$E_{2x} = E_2 \cos 0$$

$$= 3.18 \times 10^7 \text{ N/C}$$

$$\therefore E_{x_{\text{net}}} = E_{1x} - E_{2x}$$

$$= 0 \text{ N/C}$$

Y component:

$$E_{1y} = E_1 \sin \theta$$

$$= 4.5 \times 10^7 \sin 45^\circ$$

$$= 3.18 \times 10^7 \text{ N/C}$$

$$E_{2y} = E_2 \sin 0$$

$$= 3.18 \times 10^7 \text{ N/C}$$

$$\therefore E_{2y_{\text{net}}} = E_{1y} + E_{2y}$$

$$= 6.36 \times 10^7 \text{ N/C}$$

Answers to the Q. NO-08

we know that,

$$E = \frac{k \cdot q}{r^2}$$

$$\begin{aligned} \therefore q &= \frac{E r^2}{k} \\ &= \frac{5 \times (0.6)^2}{9 \times 10^9} \\ &= 2 \times 10^{-10} \text{ C} \end{aligned}$$

Here,

$$E = 5 \text{ N/C}$$

$$r = 60 \text{ cm}$$

$$= 0.6 \text{ m}$$

Ans:

Answer to the Q. NO-09

for a dipole the electric field,

$$E = 5.4 \times 10^{-8} \text{ N/C}$$

$$\therefore p = qd$$

$$= 2 \times 1.6 \times 10^{-19} \times 0.75 \times 10^{-9}$$

$$= 2.4 \times 10^{-28}$$

$$\therefore \text{Torque, } \tau = pE \sin \theta$$

a. $\rightarrow$ for parallel,

$$\begin{aligned}\tau &= PE \sin \theta \\ &= PE \sin 0^\circ \quad [\theta = 0^\circ] \\ &= 0 \text{ Nm}\end{aligned}$$

b. $\rightarrow$ for perpendicular,

$$\begin{aligned}\tau' &= PE \sin \theta \\ &= PE \sin 90^\circ \\ &= 2.4 \times 10^{-28} \times 5.4 \times 10^{-8} \\ &= 1.296 \times 10^{-35} \text{ Nm}\end{aligned}$$

Answer to the Q. NO-10

13

We know,

$$W = -\Delta U$$

$$= -(U_f - U_i)$$

$$= - \left\{ -PE \cos \theta_f + PE \cos \theta_i \right\}$$

$$= PE \cos \theta_f - PE \cos \theta_i$$

$$= -PE \left\{ \cos \theta_i - \cos \theta_f \right\}$$

$$= -3.2 \times 10^{-24} \times 56 \left\{ \cos 65^\circ - \cos 180^\circ \right\}$$

$$= 2.54 \times 10^{-22} \text{ J}$$

Here,

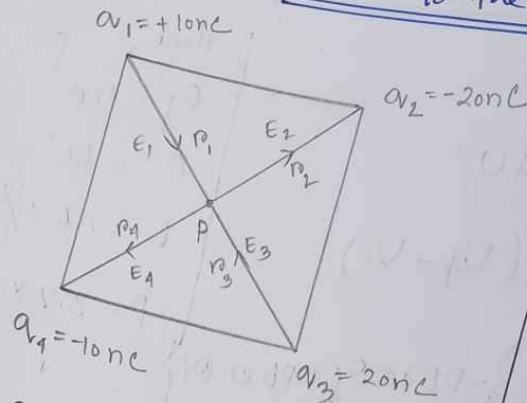
$$\theta_f = 180^\circ$$

$$\theta_i = 65^\circ$$

$$E = 56 \text{ N/C}$$

$$P = 3.2 \times 10^{-24} \text{ Cm}$$

Answer to the Q. NO-11



Here,
 Diagonal, $D = a\sqrt{2}$
 $= 0.05\sqrt{2}$
 $= 0.07\text{m}$

$$r = \frac{a\sqrt{2}}{2}$$

$$= 0.03\text{m}$$

$$r_1 = r_2 = r_3 = r_4 = r$$

$$E_1 = \frac{kq_1}{r^2} = \frac{9 \times 10^9 \times 10 \times 10^{-9}}{(0.03)^2}$$

$$= 1 \times 10^5 \text{ N/C}$$

$$E_2 = \frac{kq_2}{r^2} = \frac{9 \times 10^9 \times 20 \times 10^{-9}}{(0.03)^2}$$

$$= 2 \times 10^5 \text{ N/C}$$

$$E_3 = \frac{kq_3}{r^2} = \frac{9 \times 10^9 \times 20 \times 10^{-9}}{(0.03)^2}$$

$$= 2 \times 10^5 \text{ N/C}$$

$$E_4 = \frac{kq_4}{r^2} = \frac{9 \times 10^9 \times 10 \times 10^{-9}}{(0.03)^2}$$

$$= 1 \times 10^5 \text{ N/C}$$

$$\therefore E_{\text{net}} = E_3 - E_1 = 2 \times 10^5 - 1 \times 10^5$$

$$= 1 \times 10^5 \text{ N/C}$$

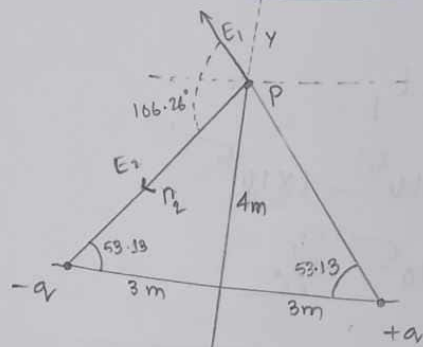
$$\begin{aligned}\therefore E_{24 \text{ net}} &= E_2 - E_1 \\ &= 2 \times 10^5 - 1 \times 10^5 \\ &= 1 \times 10^5 \text{ N/C}\end{aligned}$$

$$\begin{aligned}\therefore E_{\text{net}} &= \sqrt{E_{13}^2 + E_{24}^2 + 2 E_{13} \cdot E_{24} \cdot \cos 90^\circ} \\ &= 1.41 \times 10^5 \text{ N/C}\end{aligned}$$

$$\begin{aligned}\therefore \theta &= \tan^{-1} \left\{ \frac{E_{13} \sin 90^\circ}{E_{24} + E_{13} \cos 90^\circ} \right\} \\ &= \tan^{-1} \left\{ \frac{1 \times 10^5 \sin 90^\circ}{1 \times 10^5 + 1 \times 10^5 \cos 90^\circ} \right\} \\ &= 45^\circ\end{aligned}$$

Ans:

Answer to the Q. NO-12



$$r_1 = \sqrt{4^2 + 3^2} = 5\text{ m}$$

$$r_1 = r_2 = r = 5\text{ m}$$

$$E_1 = \frac{kq}{r_1^2} = \frac{9 \times 10^9 \times 4.80 \times 10^{-10}}{(5)^2} = 1.728 \times 10^{-10} \text{ N/C}$$

$$E_2 = \frac{kq}{r_2^2} = \frac{9 \times 10^9 \times 4.80 \times 10^{-10}}{(5)^2} = 1.728 \times 10^{-10} \text{ N/C}$$

Both E_1 and E_2 , Magnatude is same. so, their result will be at the middle of that two electric field.

Answer to the Q. NO-13

We know,

$$E = \frac{k \cdot q_e}{r^2}$$
$$= \frac{9 \times 10^9 \times 1.6 \times 10^{-19}}{(0.5 \times 10^{-10})^2}$$
$$= 5.76 \times 10^{11} \text{ N/C}$$

Here,

$$r = 0.5 \times 10^{-10} \text{ m}$$

$$e = 1.6 \times 10^{-19} \text{ C}$$

$$k = 9 \times 10^9 \text{ Nm}^2 \text{C}^{-2}$$

Answer to the Q. NO-14

a)

We know,

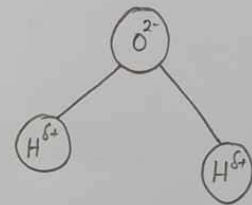
$$P = qd$$

$$\Rightarrow d = \frac{P}{q} \text{ --- (1)}$$

$$q = ne$$

$$= (8+2) \times 1.6 \times 10^{-19}$$

$$= 1.6 \times 10^{-18} \text{ C}$$



Here,

$$P = 6.2 \times 10^{-30} \text{ Cm}$$

from ①

$$d = \frac{6.2 \times 10^{-30}}{1.6 \times 10^{-18}}$$
$$= 3.875 \times 10^{-12} \text{ m}$$

b. >

$$\text{Torque, } \tau = \vec{p} \times \vec{E}$$

$$= pE \sin \theta$$

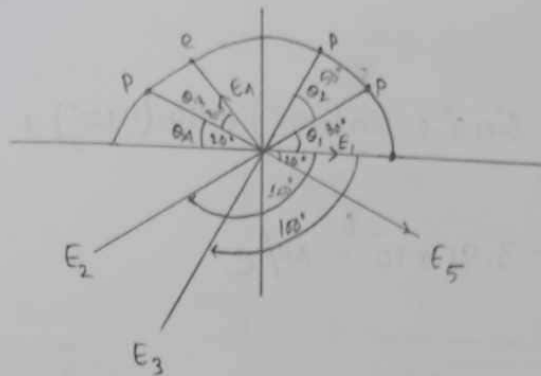
$$= pE \sin 90^\circ$$

$$= 6.2 \times 10^{-30} \times 1.5 \times 10^4 \times \sin 90^\circ$$

$$= 9.3 \times 10^{-26} \text{ Nm}$$

Ans.

Answers to the Q. NO-15



$$r = 2 \text{ cm} \\ = 0.02 \text{ m}$$

$$E = \frac{kq}{r^2} = \frac{9 \times 10^9 \times (1.6 \times 10^{-19})}{(0.02)^2} \\ = 3.6 \times 10^{-6} \text{ N/C}$$

$$\therefore E_1 = E_2 = E_3 = E_4 = E_5 = E$$

X component:

$$E_{1x} = E \cos 30^\circ$$

$$E_{2x} = E \cos 150^\circ$$

$$E_{3x} = E \cos(-100^\circ)$$

$$E_{4x} = E \cos(130^\circ)$$

$$E_{5x} = E \cos(-20^\circ)$$

Y component:

$$E_{1y} = E \sin 0^\circ$$

$$E_{2y} = E \sin(-150^\circ)$$

$$E_{3y} = E \sin(-100^\circ)$$

$$E_{4y} = E \sin(130^\circ)$$

$$E_{5y} = E \sin(-20^\circ)$$

$$\therefore E_{x_{\text{net}}} = E \left\{ \cos 30^\circ + \cos 150^\circ + \cos 100^\circ + \cos 130^\circ + \cos 20^\circ \right\}$$

$$= 9.25 \times 10^{-7} \text{ N/C}$$

$$E_{y_{\text{net}}} = E \left\{ \sin 0^\circ + \sin(-15^\circ) + \sin(-100^\circ) + \sin(130^\circ) + \sin(-20^\circ) \right\}$$

$$= -3.81 \times 10^{-6} \text{ N/C}$$

$$\therefore E_{\text{net}} = \sqrt{E_x^2 + E_y^2}$$

$$= 3.43 \times 10^{-6} \text{ N/C}$$

$$\therefore \theta_{\text{net}} = \tan^{-1} \left\{ \frac{E_{y_{\text{net}}}}{E_{x_{\text{net}}}} \right\}$$

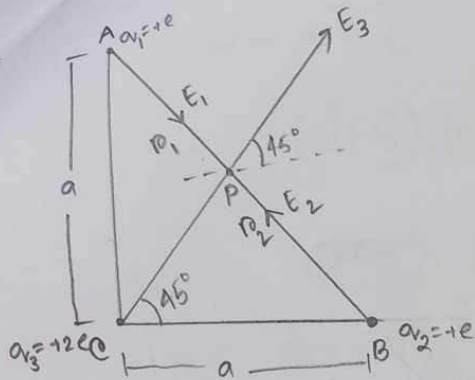
$$= \tan^{-1} \left\{ \frac{-3.81 \times 10^{-6}}{9.25 \times 10^{-7}} \right\}$$

$$= -76.4^\circ$$

$\therefore \theta$ is -76.4° clockwise from positive x axis.

Answers to the Q. NO-16

21



$$E_1 = \frac{k \cdot q_1}{r_1^2} = \frac{9 \times 10^9 \times 1.6 \times 10^{-19}}{(4.24 \times 10^{-6})^2} = 40 \text{ N/C}$$

$$E_2 = \frac{k \cdot q_2}{r_2^2} = \frac{9 \times 10^9 \times 1.6 \times 10^{-19}}{(4.24 \times 10^{-6})^2} = 40 \text{ N/C}$$

$$E_3 = \frac{k \cdot q_3}{r_3^2} = \frac{9 \times 10^9 \times 2 \times 1.6 \times 10^{-19}}{(4.24 \times 10^{-6})^2} = 160.2 \text{ N/C}$$

$$E_{12} = E_1 - E_2 = (40 - 40) \text{ N/C} = 0 \text{ N/C}$$

$$E_{\text{net}} = \sqrt{E_3^2 + E_{12}^2} = 160.2 \text{ N/C}$$

Here,

$$a = 6 \mu\text{m} = 6 \times 10^{-6} \text{ m}$$

$$q_1 = q_2 = +e$$

$$q_3 = +2e$$

$\triangle ACB$,

$$AB = \sqrt{AC^2 + CB^2} = \sqrt{(6 \mu\text{m})^2 + (6 \mu\text{m})^2} = 8.48 \times 10^{-6} \text{ m}$$

$$BP = AP = \frac{8.48 \times 10^{-6}}{2} = 4.24 \times 10^{-6} \text{ m}$$

$$CP = 4.24 \times 10^{-6} \text{ m}$$

$$\begin{aligned}\therefore E_{3x} &= E_3 \cos 45^\circ \\ &= 113.27 \text{ N/C}\end{aligned}$$

$$\begin{aligned}E_{3y} &= E_3 \sin 45^\circ \\ &= 113.27 \text{ N/C}\end{aligned}$$

$$\begin{aligned}\therefore \theta &= \tan^{-1} \left\{ \frac{E_{3y}}{E_{3x}} \right\} \\ &= 45^\circ\end{aligned}$$

$\therefore E_{\text{net}}$ is 45° about positive x axis (anticlockwise)

a.

$$U = -\vec{P} \cdot \vec{E}$$

$$= (3\hat{i} + 4\hat{j}) \cdot (1.24 \times 10^{-30}) \cdot (4000)\hat{i}$$

$$= (12000) \cdot 1.24 \times 10^{-30}$$

$$= -1.488 \times 10^{-26} \text{ J}$$

b.

given,

$$\vec{P} = (3\hat{i} + 4\hat{j}) (1.24 \times 10^{-30})$$

$$\vec{E} = 4000 \hat{i}$$

$$\therefore \vec{P} \times \vec{E} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 4 & 0 \\ 4000 & 0 & 0 \end{vmatrix}$$

$$= \hat{i}(0-0) - \hat{j}(0-0) + \hat{k}(0-16000)$$

$$= -16000 \hat{k}$$

$$|\vec{P} \times \vec{E}| = \sqrt{(-16000)^2} = 16000$$

$$\therefore W = 1.24 \times 10^{-30} \times |\vec{P} \times \vec{E}|$$

$$= 1.24 \times 10^{-30} \times 16000$$

$$= 1.984 \times 10^{-26} \text{ J}$$

Here,

$$\vec{P} = (3\hat{i} + 4\hat{j}) (1.24 \times 10^{-30})$$

$$\vec{E} = (4000 \text{ N/C}) \hat{i}$$

Answer to the Q. No - 18

a.) We know,

$$\begin{aligned} V &= V_0 + at \\ &= V_0 - \frac{Eq}{m} \times t \\ &= 40 \times 10^3 - \frac{50 \times 1.6 \times 10^{-19} \times 1.5 \times 10^{-9}}{9.11 \times 10^{-31}} \\ &= 2.67 \times 10^4 \text{ m/s} \end{aligned}$$

Here,

$$\begin{aligned} V_0 &= 40 \text{ km/s} \\ &= 40 \times 10^3 \text{ m/s} \end{aligned}$$

$$E = 50 \text{ N/C}$$

$$\begin{aligned} t &= 1.5 \text{ ns} \\ &= 1.5 \times 10^{-9} \text{ s} \end{aligned}$$

$$q = 1.6 \times 10^{-19} \text{ C}$$

$$m = 9.11 \times 10^{-31} \text{ kg}$$

b.)

$$\begin{aligned} d &= \left(\frac{V + V_0}{2} \right) t \\ &= \left\{ \frac{2.67 \times 10^4 + 40 \times 10^3}{2} \right\} \times 1.5 \times 10^{-9} \\ &= 5.00 \times 10^{-5} \text{ m} \end{aligned}$$

Ans.

Answers to the Q. NO-19

we know that,

$$F = \frac{2kp}{z^3}$$

$$= \frac{2 \times 9 \times 10^9 \times 3.6 \times 10^{-29}}{(25 \times 10^{-9})^3}$$

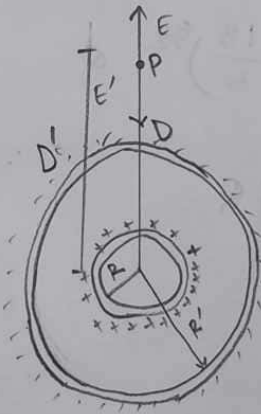
$$= 1.0368 \times 10^{-3} \text{ N}$$

Here,

$$p = 3.6 \times 10^{-29} \text{ Cm}$$

$$z = 25 \times 10^{-9} \text{ m}$$

if the dipole is oriented such that $\vec{p}$ is in the $+z$ direction, then $\vec{F}$ points in the $-z$ direction.

Answers to the Q. NO-20

$$\therefore E = E'$$

$$\Rightarrow \frac{2kQ}{(z+R')^{3/2}} = \frac{2kQ'}{(z+R')^{3/2}}$$

$$\Rightarrow \frac{Q}{(2^v + R^v)^{3/2}} = \frac{a'}{\{2^v + (3R)^v\}^{3/2}}$$

$$\Rightarrow \frac{Q}{\{(2R)^v + R^v\}^{3/2}} = \frac{a'}{\{(2R)^v + (3R)^v\}^{3/2}}$$

$$\Rightarrow \frac{Q}{(5R^v)^{3/2}} = \frac{a'}{(13R^v)^{3/2}}$$

$$\Rightarrow \frac{Q}{\cancel{(5R^v)} a'} = \frac{(5R^v)^{3/2}}{(13R^v)^{3/2}}$$

$$\therefore a' = \cancel{(13R^v)} \left(\frac{13R^v}{5R^v} \right)^{3/2} Q$$

$$= \left(\frac{13}{5} \right)^{3/2} \cdot Q$$

$$\therefore a' = 4.14 Q$$

Answers to the Q. No - 21

a.) The magnitude of electric field,

Here,

$$q = -2 \times 10^{-9} \text{ C}$$

$$F = 3 \times 10^{-6} \text{ N}$$

$$E = F/q$$

$$= \frac{3 \times 10^{-6}}{2 \times 10^{-9}}$$

$$= 1.5 \times 10^3 \text{ N C}^{-1}$$

b.)

The magnitude of electric force is,

$$F_e = qE$$

$$= 1.6 \times 10^{-19} \times 1.5 \times 10^3$$

$$= 2.4 \times 10^{-16} \text{ N}$$

c.) The gravitational force on the proton is,

$$F_g = mg$$

$$= 1.67 \times 10^{-27} \times 9.8$$

$$= 1.63 \times 10^{-26} \text{ N}$$

d.)

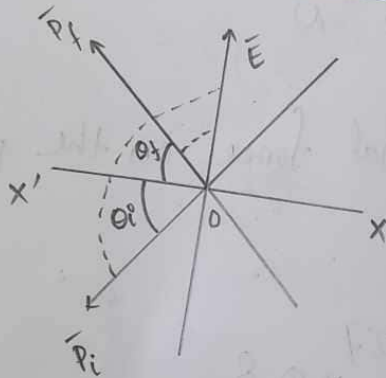
The ratio between electrostatic force and gravitational force is,

$$\frac{F_e}{F_g} = \frac{2.4 \times 10^{-16}}{1.63 \times 10^{-26}}$$

$$= 1.47 \times 10^{10}$$

Ans.

Answer to the Q. No-22



Here,

$$d = 1.2 \text{ fm} = 1.2 \times 10^{-15} \text{ m}$$

$$\theta_i = 20^\circ = 90^\circ + 20^\circ = 110^\circ$$

$$\theta_f = 20^\circ = 90^\circ - 20^\circ = 70^\circ$$

$$E = 3 \times 10^6 \text{ N/C}$$

$$P = 1.6 \times 10^{-27} \text{ C m}$$

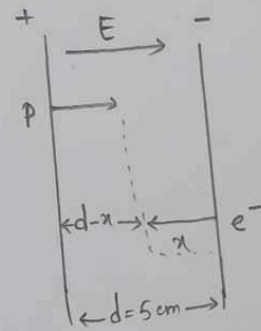
Answer to the Q. NO-23

• acceleration of electrons,

$$a_e = \frac{eE}{m_e}$$

• acceleration of protons,

$$a_p = \frac{eE}{m_p}$$



So, distance by electrons,

$$x = \frac{1}{2} a_e t^2 \quad \text{--- (i)}$$

by protons,

$$d-x = \frac{1}{2} a_p t^2 \quad \text{--- (ii)}$$

$$\text{(i)} \div \text{(ii)} \quad \frac{x}{d-x} = \frac{\frac{1}{2} a_e t^2}{\frac{1}{2} a_p t^2}$$

$$\Rightarrow \frac{x}{d-x} = \frac{a_e}{a_p}$$

$$\Rightarrow \frac{x}{d-x} = \frac{eE/m_e}{\frac{eE}{m_p}}$$

$$\Rightarrow \frac{x}{d-x} = \frac{m_p}{m_e}$$

a)

we know,

$$\Delta U = -\vec{p} \cdot \vec{E}$$

$$= U_f - U_i$$

$$= p_i E_i \cos \theta_i - p_f E_f \cos \theta_f$$

$$= pE (\cos \theta_i - \cos \theta_f)$$

$$= 1.6 \times 10^{-27} \times 3 \times 10^6 \left\{ \cos 110^\circ - \cos 70^\circ \right\}$$

$$= 3.28 \times 10^{-21} \text{ J}$$

b)

$$\Delta U = q \Delta V$$

$$\therefore \Delta V = \frac{\Delta U}{q} \quad \text{--- (1)}$$

$\therefore$ Dipole moment, $p = qd$

$$\Rightarrow q = \frac{p}{d}$$

$$= \frac{1.6 \times 10^{-27}}{1.2 \times 10^{-15}}$$

$$= 1.33 \times 10^{-12}$$

from (i)

$$\Delta V = \frac{\Delta U}{q}$$

$$= \frac{3.28 \times 10^{-21}}{1.33 \times 10^{-12}}$$

$$= 2.46 \times 10^{-9} \text{ volt.}$$

$$\Rightarrow x m_e = d m_p - x m_p$$

$$\Rightarrow x m_e + x m_p = d m_p$$

$$\Rightarrow x = \frac{d m_p}{(m_e + m_p)}$$

$$\Rightarrow x = \frac{5 \times 1.67 \times 10^{-27}}{(9.11 \times 10^{-31} + 1.67 \times 10^{-27})}$$

$$\therefore x = 4.9973 \text{ cm}$$

$$\text{from the plate} = d - x$$

$$= 5 - 4.9973$$

$$= 2.72 \times 10^{-3} \text{ cm}$$